

GAIRA

Hierarchical Cross-Modal Validation (V4)

Null-calibrated recoverability across Raman, Ag-SERS, controlled perturbation and biological matrix. Every representation metric is calibrated against an analyte-mismatched null; recovery is defined statistically, never from a raw cosine threshold.

Frozen atlas fingerprint 09ed804a40836f4a05a91ba10900cded

51 matched pure analytes · reproduces V3 bit-for-bit · SERS validates, never trains

Recovered analyte-specifically: latent 7/51 · MSS 3/51 · theme 4/51 · perturbation 3/51 · matrix 9/51 (serum-tested). Raw cosines (MSS 0.74, theme 0.92) are shared background.

1 · Executive summary

Raman → Ag-SERS transfer cannot be described by one score — and V4's central result is that it must not be described by one **level** either. Calibrated against an analyte-mismatched null, **analyte-specific recovery is rare at every level**.

The high raw cosines are almost entirely shared background: MSS matched 0.740 vs null 0.732 (separation 0.008); theme 0.918 vs 0.915 (0.002). A high raw cosine is broad biochemical interpretation, **not** analyte identity.

MSS is **not** the primary cross-modal metric — its null separation is smaller than the latent coordinates', and its 3 recovered analytes are a strict subset of the 7 recovered at the latent level. The strongest evidence is functional perturbation (3 analytes). The purine attractor appears in the unspiked-serum blank (purine 0.27) before any analyte is added.

The GAIRA representation hierarchy (null-calibrated)

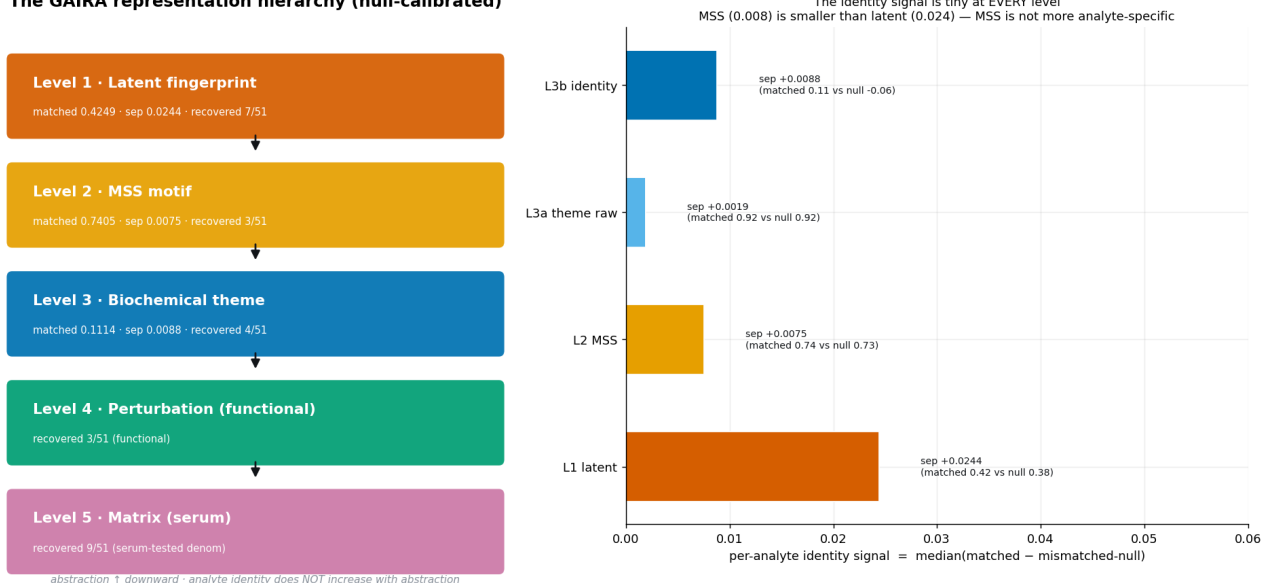


Figure 1. The null-calibrated hierarchy. The per-analyte identity signal (matched – mismatched null) is tiny at every level and smallest for MSS.

2 · How many analytes are recoverable at each level?

Level	n	denom	frac	CI low	CI high
latent	7	51	0.137	0.059	0.235
MSS	3	51	0.059	0.000	0.137
theme	4	51	0.078	0.020	0.157
top3_over_null	28	51	0.549	0.412	0.686
argmax_robust	13	51	0.255	0.137	0.373
perturbation	3	51	0.059	0.000	0.137
matrix	9	51	0.176	0.078	0.294

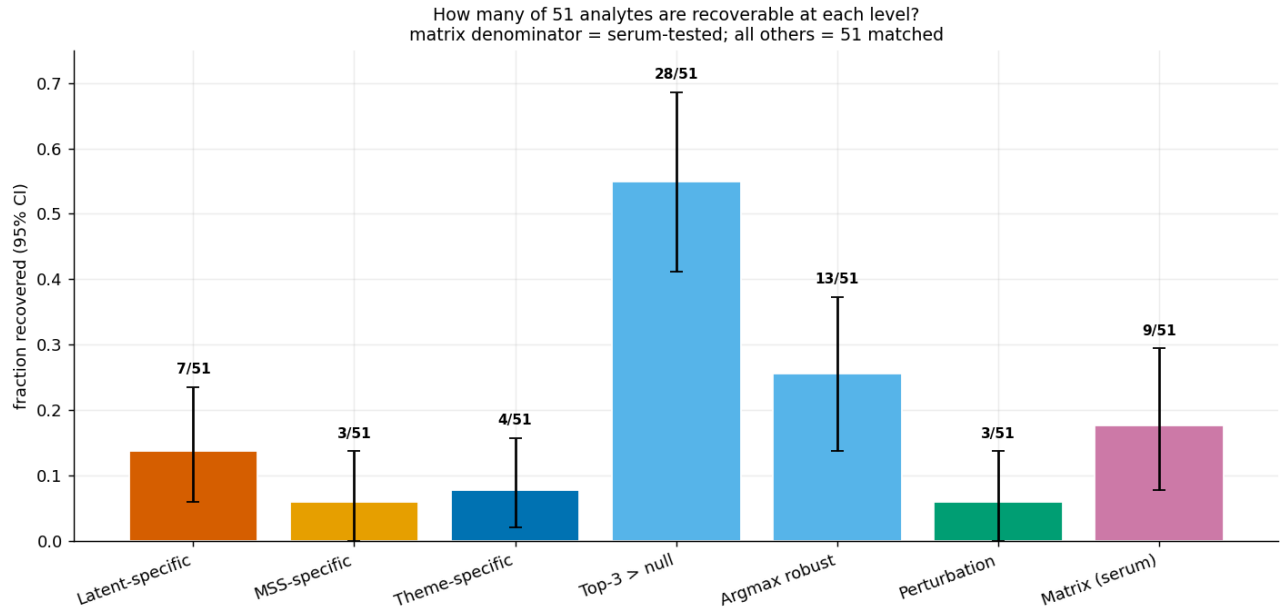


Figure 2. Fractions with 95% CI. Matrix denominator = serum-tested; all others = 51 matched.

Recovered analyte names (rank-1 + jackknife-stable):

- **latent** (7): creatinine, glutathione, hypoxanthine, thymine, urate, urea, xanthine
- **MSS** (3): creatinine, urea, xanthine
- **theme** (4): creatinine, hypoxanthine, urea, xanthine
- **perturbation** (3): adenine, ergothioneine, urate
- **matrix** (9): adenine, albumin, ergothioneine, guanine, hydroxyproline, hypoxanthine, oleate, urate, xanthine

3 · Per-analyte recovery matrix

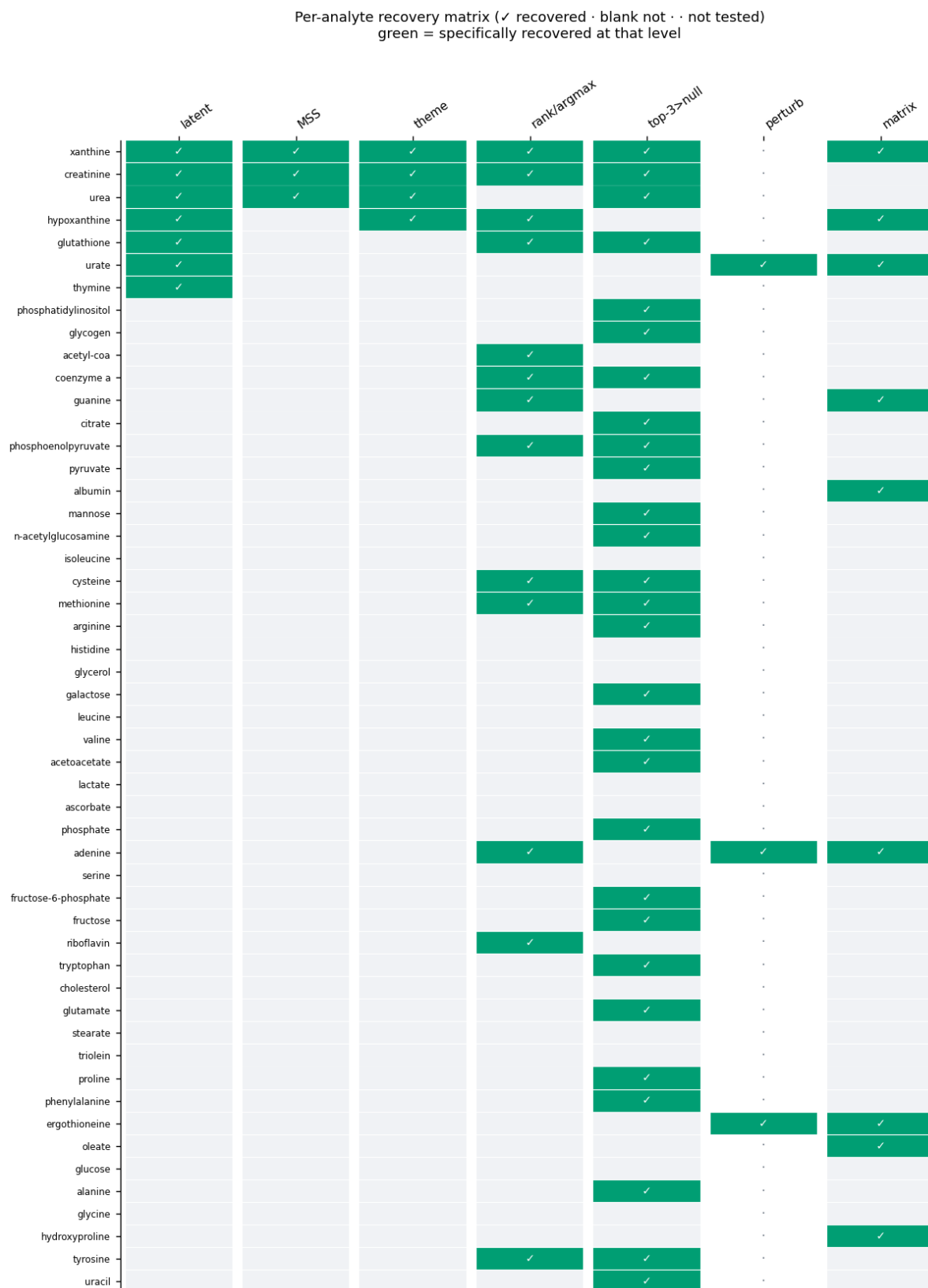


Figure 3. ✓ recovered · blank not · dot not-tested. Strong chemisorbers dominate; most analytes are broad-theme only.

4 · Matched vs mismatched — which metric carries identity?

Matched vs mismatched distributions — overlap = the metric does NOT carry analyte identity

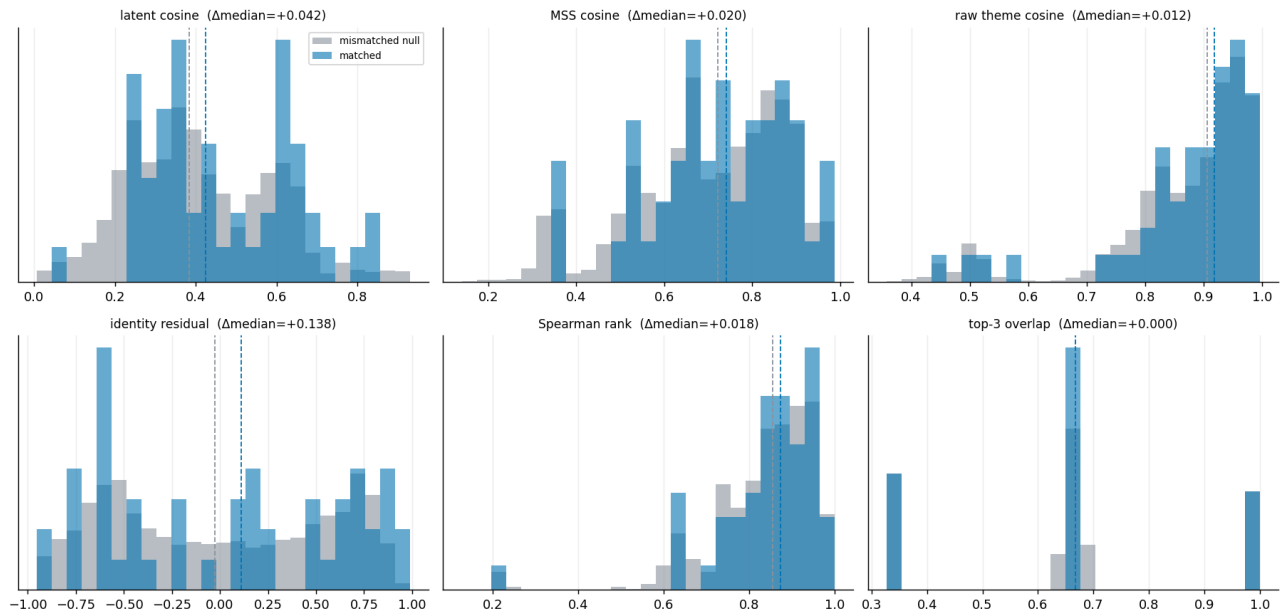


Figure 4. Matched (blue) and mismatched-null (grey) distributions overlap heavily at every level — the metrics carry little analyte identity.

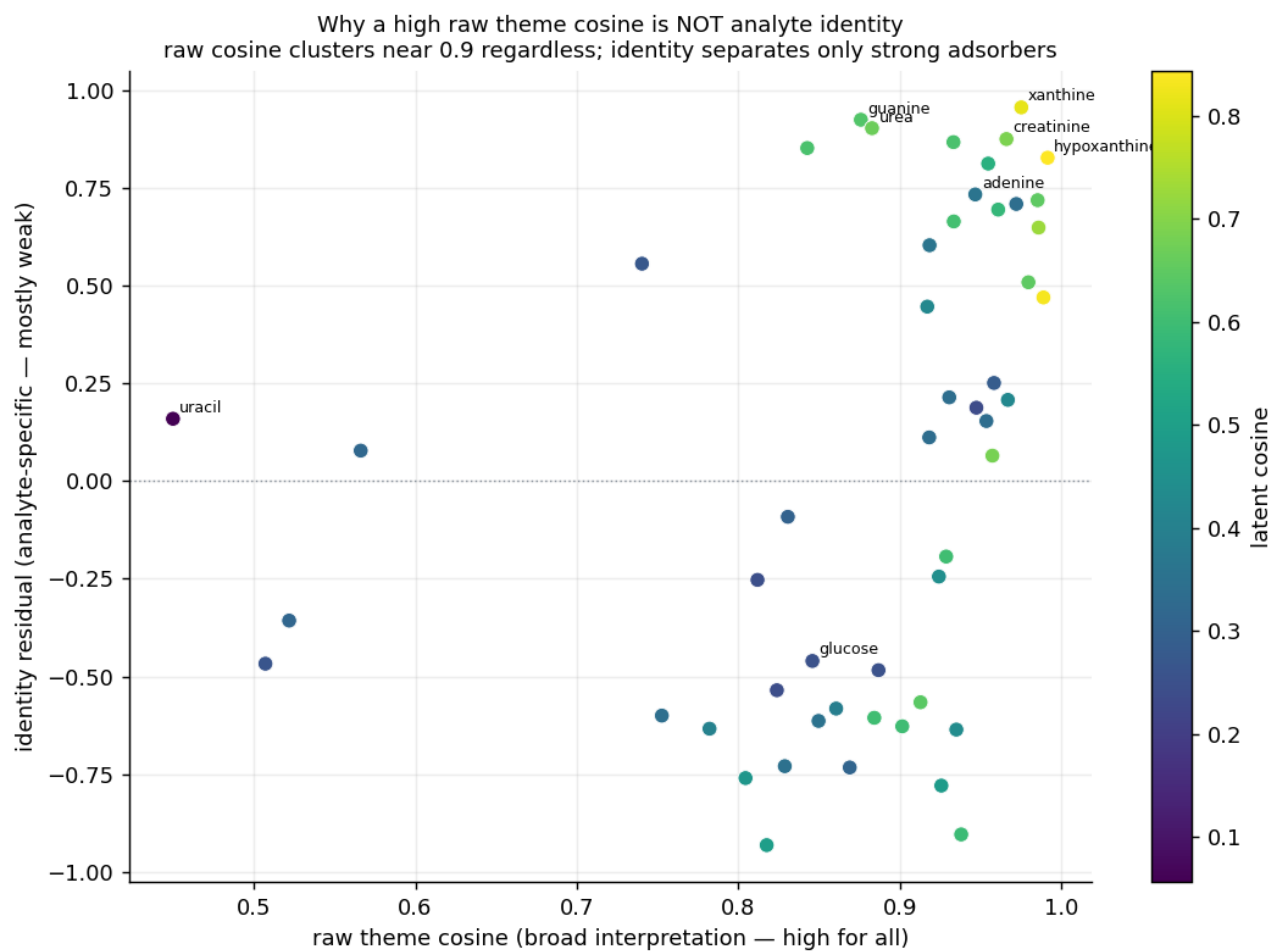


Figure 7. Raw theme cosine clusters near 0.9 for all analytes; the identity residual separates only the strong adsorbers.

5 · MSS is not primary (tested, not assumed)

MSS motif cosine (0.740) looks like a robust middle layer, but its mismatched null is 0.732 — the 0.74 is background. Only 3/51 clear the null, all already latent-recovered.

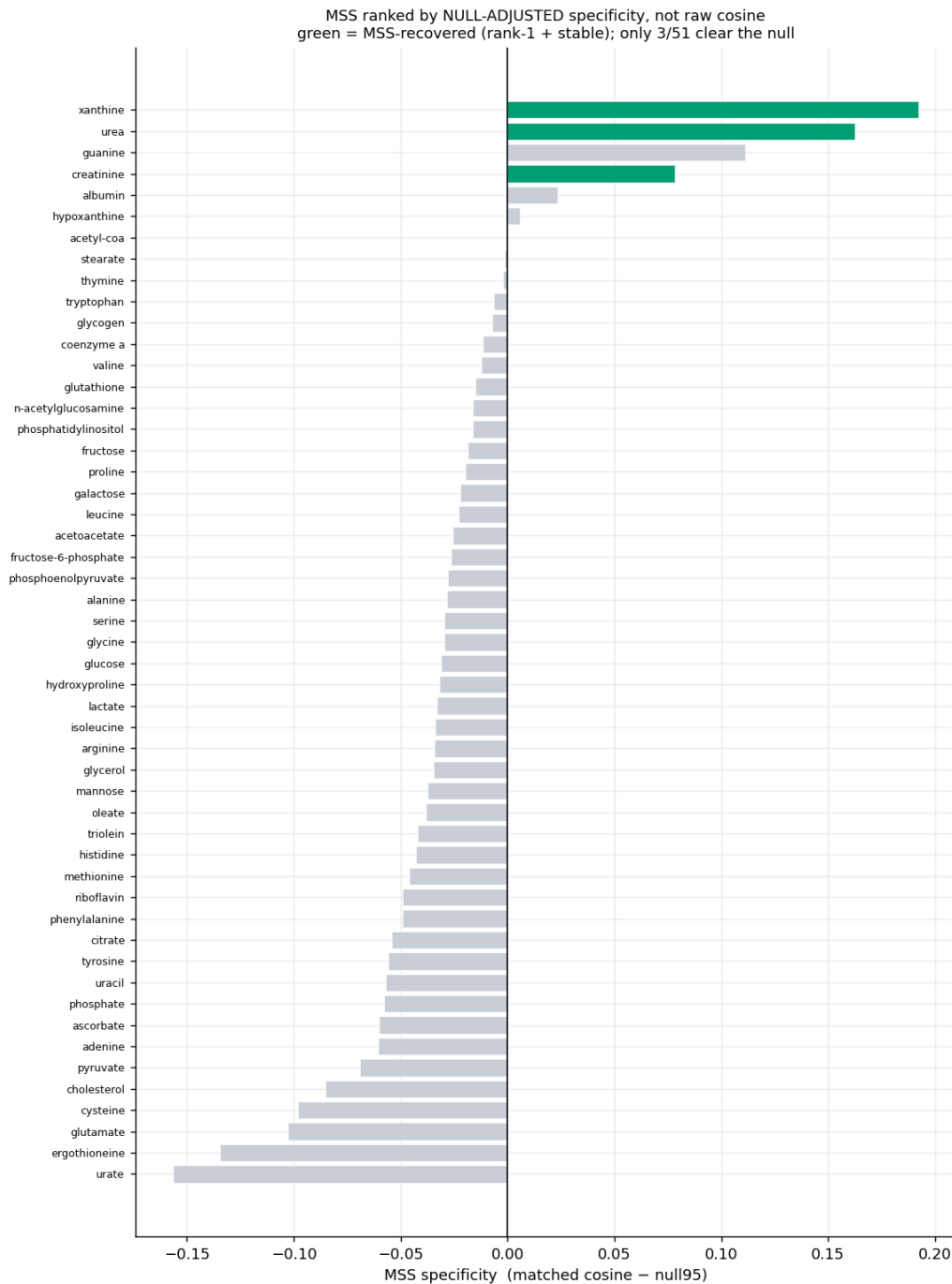


Figure 6. MSS ranked by null-adjusted specificity (matched - null95), not raw cosine. Only 3 clear the null.

Level	matched	null	separation	rank-1	recovered
L1 latent	0.425	0.380	0.024	7	7
L2 MSS	0.741	0.732	0.007	3	3

Level	matched	null	separation	rank-1	recovered
L3a theme raw (diagnostic)	0.918	0.915	0.002	3	3
L3b Raman-centered identity residual	0.111	-0.064	0.009	4	4

6 · The purine attractor — background control

The unspiked-serum-on-Ag blank projects to purine share **0.2659** with **nucleic_purine** dominant — the attractor is present before any analyte. Δ purine anti-correlates with latent ($r=-0.38$, $p=0.006$) and MSS ($r=-0.40$, $p=0.003$). No pure Ag-colloid buffer blank exists, so the mechanism is not fully isolated: we describe the attractor as phenomenological, not as Ag binding alone.

The purine attractor — phenomenological, present in the background BEFORE analyte addition

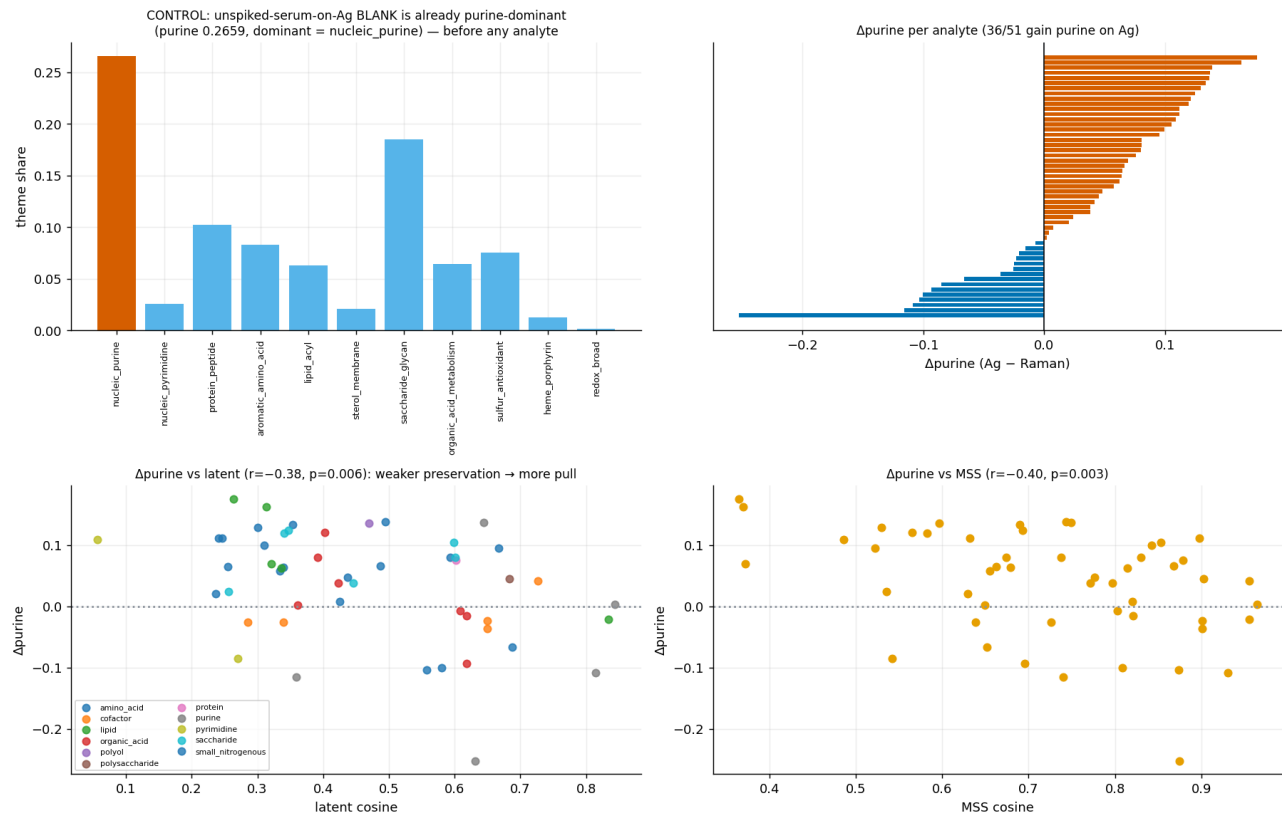


Figure 8. The attractor and its controls.

7 · Perturbation & matrix

Level 4 — functional perturbation validation (ONLY adenine, ergothioneine, uricase)

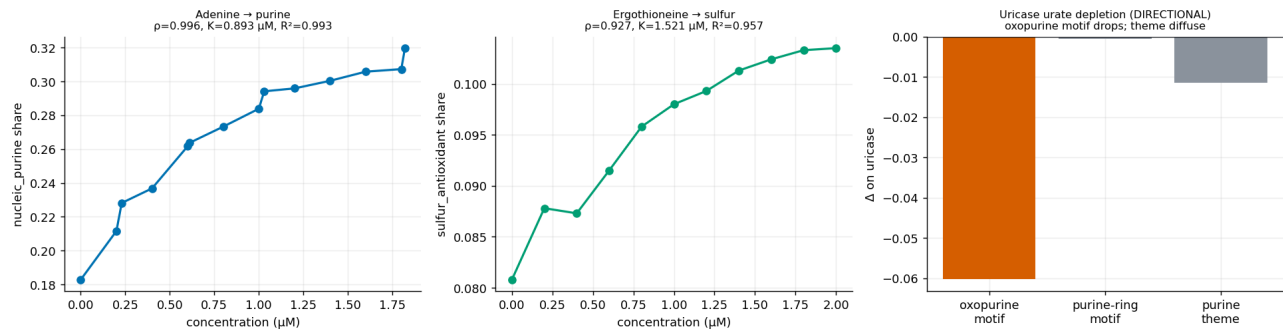


Figure 9. Functional validation for the only three tested analytes (adenine, ergothioneine dose; uricase directional).

Matrix: no pure metric significantly predicts serum displacement; only overall confidence correlates ($r=0.71$, $p=5e-9$), likely reflecting signal strength, not identity.

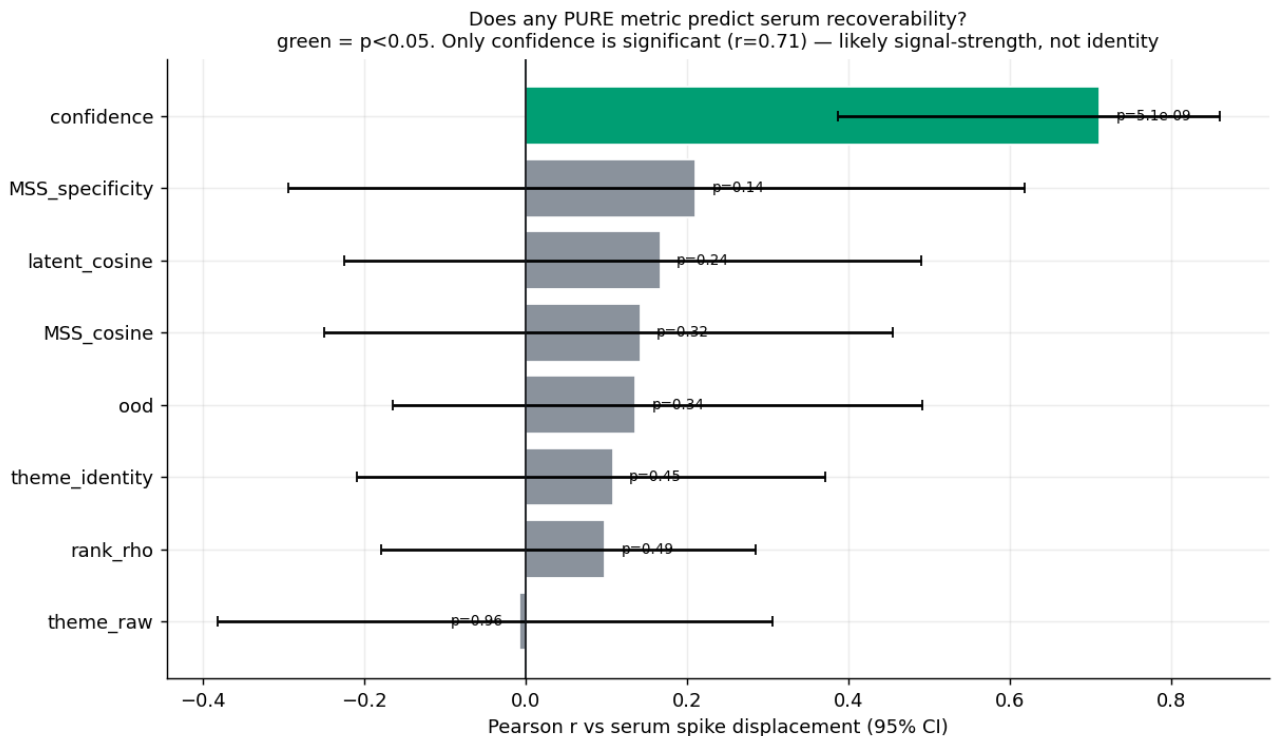


Figure 10. Pure-metric predictors of serum recovery with 95% CI. Only confidence is significant.

8 · Representative analytes

Representative analytes: spectrum → latent → themes → identity residual → evidence (blue = Raman, red = Ag-SERS)



Figure 11. Spectrum → latent → themes → identity residual → evidence for seven representative analytes (blue Raman, red Ag-SERS).

9 · Metric-selection decision table

Decided by the null analyses, not prior preference:

Purpose	Primary metric	Why (null result)
Exact spectral / substrate fidelity	Latent cosine	strongest cosine identity: 7/51 rank-1, sep 0.024
Cross-modal chemical-motif preservation	Latent cosine (MSS supporting)	MSS NOT primary: sep 0.0075, 3/51 = strict subset of latent's 7
Broad biochemical interpretation	Raw theme BSV / radar	high (0.92) but matched approx equals null - interpretation, not identity
Analyte-specific theme retention	Raman-centered identity residual	null-adjusted; 4/51 rank-1
Gross theme ordering	Spearman rho	matched approx null - descriptive only
Strict dominant category	Argmax + top-two margin	brittle; 13/51 robust
Functional biochemical response	Perturbation sensitivity	strongest evidence; 3 analytes
Biological-mixture visibility	Matrix recoverability	separate property; 9/51 serum-strong

Verdict: latent cosine is the best cross-modal identity cosine (7/51); MSS is supporting, not primary; raw theme and Spearman are broad-interpretation descriptors; functional perturbation is the strongest evidence; matrix is a separate property.

10 · Limitations & conclusions

Limitations. Raman-trained atlas; no learned modality correction; purine mechanism not fully isolated (no buffer blank); 3 perturbation cases; recovery depends on null thresholds and the discrete retrieval p-floor (1/51, so BH-FDR is degenerate at $N=51$); confidence $\neq$ analyte identifiability; incomplete Au-SERS grounding; 5 replicates/analyte.

Conclusions. Cross-modal biochemical recovery is hierarchical and rare: only the strong-chemisorber minority retains analyte-specific latent structure (7/51); motif and theme identity are weaker (3–4/51); raw cosines are broad-interpretation background; the purine attractor is a background phenomenon; functional perturbation (3 analytes) is the strongest evidence. GAIRA reports the whole calibrated hierarchy and never calls an analyte 'detectable' from a raw cosine.

Provenance. Frozen atlas 09ed804a40836f4a05a91ba10900cded verified unchanged; matched values reproduce V3 (max abs diff 0.0); deterministic; reruns identical. Full method in METRICS_AND_DECISION_RULES.md; source audit in AUDIT_OF_V3_METRICS.md.